

New Frontiers of Human-centered Explainable AI (HCXAI): Participatory Civic AI, Benchmarking LLMs, XAI Hallucinations, and Responsible AI Audits

Upol Ehsan
Northeastern University
Boston, Massachusetts, USA
Harvard University
Boston, Massachusetts, USA
u.ehsan@northeastern.edu

Elizabeth A Watkins
Intelligent Systems Research (ISR)
Intel Labs
Santa Clara, California, USA
elizabeth.watkins@intel.com

Philipp Wintersberger
Intelligent User Interfaces Group
Interdisciplinary Transformation
University Austria
Linz, Austria
philipp.wintersberger@it-u.at

Carina Manger
Human-Computer Interaction Group
Technische Hochschule Ingolstadt
Ingolstadt, Germany
carina.manger@thi.de

Nina Hubig
Interdisciplinary Transformation
University (IT:U)
Linz, Austria
nina.hubig@it-u.at

Saiph Savage
Civic A.I. Lab
Northeastern University
Boston, Massachusetts, USA
s.savage@northeastern.edu

Justin D. Weisz
IBM Research
Yorktown Heights, New York, USA
jweisz@us.ibm.com

Andreas Riener
Human-Computer Interaction Group
Technische Hochschule Ingolstadt
Ingolstadt, Bavaria, Germany
andreas.riener@thi.de

Abstract

Explainable AI (XAI) is *more* than just “opening” the black box — *who* opens it matters just as much, if not more, as the ways of opening it. Human-centered XAI (HCXAI) advocates that algorithmic transparency alone is not sufficient for making AI explainable. In our *fifth* CHI workshop on Human-Centered XAI (HCXAI), we shift our focus to new, emerging frontiers of explainability: (1) participatory approaches toward explainability in civic AI applications; (2) addressing hallucinations in LLMs using explainability benchmarks; (3) connecting HCXAI research with Responsible AI practices, algorithmic auditing, and public policy; and (4) improving representation of XAI issues from the Global South. We have built a strong community of HCXAI researchers through our workshop series whose work has made important conceptual, methodological, and technical impact on the field. In this installment, we will push the frontiers of work in HCXAI with an emphasis on operationalizing perspectives sociotechnically.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models.**

ACM Reference Format:

Upol Ehsan, Elizabeth A Watkins, Philipp Wintersberger, Carina Manger, Nina Hubig, Saiph Savage, Justin D. Weisz, and Andreas Riener. 2025. New Frontiers of Human-centered Explainable AI (HCXAI): Participatory Civic AI, Benchmarking LLMs, XAI Hallucinations, and Responsible AI Audits. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3706599.3706713>

1 Introduction

With its roots in expert systems from the 1980s [19] and its resurgence around 2016 with the DARPA XAI Challenge [14], Explainable Artificial Intelligence (XAI)—a field concerned with developing techniques, concepts, and processes that can help stakeholders understand the reasons behind the AI system’s decision-making [9, 15]—has come a long way. Even compared to the state of the field in 2021, the first installment of the Human-centered Explainable AI (HCXAI) workshop at CHI [11], the XAI landscape has substantially matured. The technology has matured; for instance, techniques that were experimental have been productized (e.g., IBM’s AI Explainability 360 and Microsoft’s InterpretML). Similarly, the regulatory landscape also shows a demand for XAI to foster Responsible AI (RAI). For instance, the EU AI Act and the US AI Bill of Rights all call for making AI systems explainable. Moreover, the era of Large Language Models (LLMs) has ushered a new set of problems that were previously a non-issue in XAI (e.g., hallucinations or confabulations), and needs for explaining behaviors of physical systems like automated vehicles [20] or robots [18] emerged.

Ever since its inception in 2020 [7], the field of Human-centered XAI (HCXAI) has also matured in tandem with the broader discourse in XAI and RAI. HCXAI encapsulates the philosophy that

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3706713>

not everything that is important lies inside the black box of AI. Critical insights can lie outside it *because that's where the humans are*. HCXAI argues that XAI is *more* than just “opening” the black-box—who opens it matters just as much, if not more, as the ways of opening it [7]. Given that real-world AI systems exist in sociotechnical settings, it takes more than just algorithmic transparency to make them explainable [12]. Increasing the aperture of XAI to include sociotechnical and human factors can help us focus on the most important part: *who* the human(s) is (are), what they are trying to achieve in seeking an explanation, and how to design XAI techniques that meet those needs. Given that XAI is as much of HCI's problem as it is AI's, CHI continues to be the venue to address the human side of XAI.

With four consecutive installments, the HCXAI workshop series is one of most well-attended one at CHI—over 400 researchers, practitioners, policy makers from over 19 countries spanning 9 sectors have joined us since 2021. We propose the **fifth installment** for HCXAI for two main principled reasons—*intellectual* and *communal*.

From an *intellectual* perspective, the rapidly evolving regulatory and technological landscape has created new frontiers. There is a **need to expand HCXAI** into un-or-underexplored areas like *benchmarking explainability in LLMs, participatory XAI techniques, Civic AI and Policy, XAI auditing amidst regulations, tackling hallucinations in explanations from generative AI systems, and XAI issues particularly from the Global South* (XAI has a Global North epistemic bias). Beyond these new areas, the time is ripe to revisit foundational topics such as XAI evaluation (2021), dark patterns in XAI (2022) to assess the progression since then.

There is also a *strong communal demand*. CHI has grown significantly over the past several years. Such growth is a double-edged sword - while CHI has become more inclusive to more people, the sheer number of presentations, panels, and papers can become overwhelming and difficult to navigate, especially for young scholars. It is important to provide stable footing and a welcoming presence if we hope that young entrants will make CHI their home venue and return year after year. Over the last year, we have received over 70 messages from participants who have strongly demanded that their “home community continues at CHI 2025” (Participant, XAI Researcher, Latin America), a sentiment aptly expressed by this participant:

Having attended every single workshop, I've grown to love this community... even when we've had over 100 participants, the discussions have always been intimate and personal...can't wait for the next one! (HCI researcher, South Asia)

In this 5th edition, we build on the groundwork of the first four workshops in 2021, 2022, 2023, and 2024 [10–13]. In 2021, the conversation was centered around the pressing need to systematically understand XAI evaluation methods. In 2022, the community increased in diversity and depth, and the conversation matured to address XAI harms and devise mitigation strategies. In 2023, there was a “turn to the infrastructure” with deep discussions around how we should focus on different stages of the AI's lifecycle and leverage “seamful information” [6] to augment XAI. In 2024, there was a critical reflection of whether “opening the black box” is still a

realistic goal for XAI in the era of LLMs. In 2025, for the 5th workshop, we continue to serve as a junction point for cross-disciplinary stakeholders to tackle HCXAI at the *conceptual, methodological, and technical* levels. The goals are to (1) extend the critically constructive dialogue around *operationalizing* human-centered perspectives in XAI and (2) further, expand and foster a supportive HCXAI community of diverse stakeholders.

2 Human-centered XAI: Background and Opportunities

Over the past several years, we have built a richly cross-functional community largely through our use of inclusive, social-science-informed analytic frameworks. Science and Technology Studies provides the lens of “Social Construction of Technology” (SCOT) [5], a framework that recognizes that the function and operation of technology emerges from the interplay between technologies, humans, and organizational/societal-level systems. Using such a lens enables us to allocate attention to actors working from disparate positions in this interplay. For example, one motivating principle across the history of HCXAI workshops has been to examine the “who” of explainability: how diverse *relevant social groups* (including stakeholders such as researchers, developers, or policymakers) draw on explainability's *interpretive flexibility* (meaning, the flexibility to support multiple concurrent diverging interpretations). This work takes on ever-greater urgency, as more stakeholders are impacted by, and participate in, the construction of what explainability means, how it is defined, for whom, and for what. For example, terms such as explainability, interpretability, and transparency, have been used interchangeably across distinct communities [1, 2, 17]. In addition to the questions around “by who?” and “for whom?” we ask also *why* [7] explanations are built, as explanations are deployed to serve a range of purposes including *causality, fairness, accessibility, trustworthiness, and interactivity* [2].

Any effort towards deriving value from explanations demands identifying and addressing ambiguities around definitions, audiences, and evaluation methods. This approach enables us to make drive the field and its outputs forward at three levels - conceptual, methodological, and technical.

3 Goals of the Workshop & Areas of Interest

The goals of this workshop are to operationalize concepts and frameworks in HCXAI, as the field of AI matures, develops, and in some corners explodes in novel directions. Multiple corners of the AI landscape have reached new phases of development. In policy spaces, AI audit and assessment mandates continue to grow, for example in the EU AI Act [4] and the White House executive order [16]. In the fast-paced world of GenAI, benchmarking has emerged and taken on widespread significance, even while the growth of foundation models has introduced new questions about how to evaluate foundation model capabilities versus post-training contexts of deployment. On top of these developments, agentic AI patterns have emerged, bringing with them questions about how we might run evaluations on systems which can be equipped to take (potentially irrevocable or harmful) actions, even series of actions even as human oversight and evaluation mechanisms still remain largely theoretical.

As all of this maturation takes place, questions still remain about a number of issues that human-centered XAI has been asking from the beginning: how might the field function as an inclusion opportunity, to better recognize the stakes and role of the global majority? What influence has HCXAI had on issues of bias and dark patterns in algorithmic and AI-based decision-making systems? How might we tackle dark patterns in XAI and explainability pitfalls [8]? As we think about auditing, assessment, evaluation, and benchmarking, how might we better understand how these practices function as mechanisms for larger goals of accountability, engineering improvement, and human oversight? We invite papers that both address novel opportunities for applying HCXAI methods and principles, as well as those that revisit questions have long motivated the HCXAI community.

The following list of guiding questions is *not* an exhaustive one; rather, it is provided as a source of inspiration:

- How has the concept of explainability evolved in the era of large language models (LLMs)?
- Does traditional LLM benchmarking address the needs of XAI systems? If yes, how? If not, why?
- What XAI techniques can help users detect hallucinations in LLM outputs? How can XAI techniques help users calibrate their trust in LLM outputs?
- Considering different “whos,” how should XAI be utilized for Civic AI design for policy makers and governments?
- How can inappropriate explainability methods hinder our ability to detect hallucinations in AI models?
- What role should XAI play in promoting responsible AI practices in the context of LLMs?
- What are the best practices for auditing XAI systems, particularly those incorporating generative AI?
- Which participatory design techniques can be used to fine-tune LLMs for generating personalized explanations?
- How might seamless XAI design be leveraged to red-team and stress-test LLMs?
- What is the role of XAI in the field of AI audits, especially as regulatory landscapes evolve around government use of AI?
- In civic AI contexts, how could XAI help artists protect their work against AI plagiarism and support copyright disputes, while ensuring that public cultural resources are respected?
- How might XAI systems empower gig workers and data workers and empower them to seek recourse in cases of disputes and promoting accountability?
- How can we address specific XAI challenges faced by the Global South (Majority World) ensuring that these solutions are inclusive and culturally sensitive?

4 Workshop Logistics

Pre-Workshop plans: Our pre-workshop plans serve three goals: **advertising** (to raise awareness and receive strong submissions), **building community**, and **recruiting speakers & expert reviewers**. To achieve these goals, we will use effective strategies that have a proven track record from previous events. **First**, for advertising, we will use an integrated advertising strategy that has two components - social media and mailing lists. The organizing committee has shared membership across many relevant disciplines

like HCI, AI, NLP, Sociology, Psychology, and Public Policy. We will primarily use Twitter and LinkedIn to advertise leveraging our network of over 60,000 combined followers. Beyond social media, we will distribute the Call for Papers through mailing lists (e.g., CHI, IUI, NeurIPS, AAAI). **Second**, for community building, we will use two avenues: our existing online community on Discord and social media. We are fortunate to have a thriving online community on Discord, which started and continued from the first HCXAI workshop, and which we will continuously foster to raise engagement. We will encourage community-driven activities from ex-participants to engage with prospective participants. In addition to Discord, we plan to utilize participation through social media advertisements. **Third**, as in previous years, we will recruit a Program Committee (PC) to handle at least 50-60 submissions (based on prior data) and recruit thought leaders as keynote speakers. We have a successful track record of recruiting thought leaders from both inside and outside XAI. In 2021, we had Tim Miller, a key voice in XAI. In 2022, Tania Lombrozo, whose work in the psychology of explanations is seminal to XAI, joined us. In 2023, we had a lively debate from different disciplines—Andres Paez (philosopher), Tania Lombrozo (psychologist), and Tim Miller (computer scientist). In 2024, Kush Varshney (IBM Fellow) and Janet Haven (Director, Data & Society) discussed tech to policy translation. For 2025, we already have keynote speakers from diverse threads (e.g., AI ethics, policy) expressing interest to join us. In all our efforts, we will prioritize diversity of perspectives and representation in an effort to make the workshop as accessible and equitable as possible.

Workshop Mode: Hybrid Event. HCXAI 2025 will be held as a hybrid event. The committee has a combined experience of organizing over 30 workshops and conferences [3]. We will use our experience to avoid common pitfalls and leverage the strengths of both worlds. We have introduced equity in virtual participation in previous hybrid workshops we have organized by taking a *virtual-first* design approach (as opposed to the usual *in-person first* approach). This worked tremendously well for CHI 2024 where the room had limited infrastructural support. For instance, everyone logs into Zoom regardless of whether they are in-person or not. This way, virtual participants do not feel alienated and can enjoy virtual co-presence with everyone. When people present, instead of pointing a camera at the stage where screen legibility suffers, participants share their screens on Zoom and present. As everyone is on Zoom, there are no A/V issues in the room and automated closed captioning provides essential *accessibility*. To make up for time zone differences, we also upload materials ahead of time so that participants can watch them asynchronously and submit questions ahead of time. As we have done in prior years, we will have a backup WiFi hotspot with us to give us coverage in case of internet issues. These strategies will allow us to streamline a comfortable experience for all participants. In the past, they have allowed us to *broaden participation* and make the event *globally accessible* to participants who are not able to make in-person for a host of reasons (monetary, health, visa, etc.). As in previous years, we expect around **100-125 participants** (40-50 in-person, the rest virtual). Given our track record, we are confident in facilitating in-depth discussions at this scale. *In the event CHI moves online, given our past experience, we are also fully capable of hosting the workshop online.*

Website, Publishing Workshop Proceedings, Discord Server, and Asynchronous Engagement: Our **website** (<https://hcxai.jimdosite.com/>) provides a rich source of information—all proceedings and videos are archived there. It has served as a *key portal for increased community engagement beyond the workshop*. At the time of proposal submission, this website hosts content from 2021 to 2024, which will be updated for 2025 (provided acceptance). Beyond the website, we have set up a **Discord server** that serves as an on-line space for discussions before, during, and after the workshop. Online participants will be able to access the provided activities via Discord. Taken together, **the website and the Discord server afford effective asynchronous engagement**. In the past, participants have used Discord to engage asynchronously in discussions or catch up on missed presentations using the website.

5 Workshop Structure

The workshop is planned as a full-day event consisting of **two 4-hour sessions (including coffee breaks)** (Table 1 outlines the key activities). Tentatively, the sessions will run from 9am-1pm and 2pm-6pm local time in Japan. Given CHI'25 time is ahead of most places, we will have to account for a wide range of hybrid participation options (i.e., the US West Coast lagging 16 hours behind). Thus, we plan to integrate participants in different locations independently and asynchronously. As in previous installments, the introduction of workshop participants will be mainly done in our Discord servers, while we will directly record our main events (i.e., keynote, paper presentations) so that hybrid participants from Europe and Africa can stream the session in their morning. For group activities, we will combine semi-structured discussions (following popular brainstorming techniques) with networking possibilities. Videos of poster presentations will be made available already the day before the official start of the event in Japan. Thus, while the in-person core of the workshop is scheduled as the full-day activity at CHI, the hybrid, asynchronous participation possibilities will cover a time span of 24 hours. We briefly outline the course of events for four types of attendees (on-site, remote from Europe, Asia, and the US) which will allow participants from all over the globe to engage equitably:

- **On-site and from Asia:** Participants can experience the entire event real-time: keynote, followed by first two paper session, lunch break, followed by afternoon session with paper and poster sessions and group activity.
- **From Europe:** Participants can review poster presentations the evening before the workshop and watch the keynote stream during the on-site lunch break. They can join for the afternoon session and group works
- **From the Americas:** participants can join the entire event online during their afternoon and late evening, taking advantage of online materials like other remote participants.

Two weeks before the start of the workshop, we will share reading materials (e.g., past proceedings and recent impactful HCXAI papers). We will also ramp up social media engagement using our #HCXAI hashtag. Through a dedicated Discord channel, participants will have a chance to introduce themselves and begin engaging with each other. Since online and hybrid events struggle with instantaneous rapport building, prolonging the introductory

phases has shown to be effective in promoting conversations. If the previous submission volume holds, we will have *three tracks*—full presentations (about 33%), rapid-fire poster presentations (about 67%), and the creative short-form videos-only track (“The Sanities & Insanities of XAI: Provocations & Evocations”). Presentations happen in Zoom while the discussion happens on dedicated channels in Discord. This combination promoted a smooth experience (without cluttering the chat on Zoom calls) and asynchronous engagement—speakers appreciated being able to continue the conversation on Discord even after their talks are over.

In **Session 1**, we will begin with a *brief introductory session* that aligns participants with the workshop goals, outlines key activities, and introduces the organizers. This is followed by a first networking and discussion section in form of a ‘*Happy Hour*’ where both on-site and online participants can network and engage in fun activities (like at-home scavenger hunts). Next, we will have a *fireside chat (interactive keynote)* with thought leaders from diverse disciplines like AI, HCI, policy, and regulations (e.g., Janet Haven, Kush Varshey, Tim Miller, Tanja Lombrozo). Instead of a keynote monologue, past experience suggests that an interactive fireside chat format (15-min presentation with 45-min Q&A) simultaneously facilitates engagement from the audience while reducing the speaker’s burden of preparing a long presentation. The rest of the session will include presentations of papers and posters. We will have breaks between sessions to reduce fatigue. We plan to host these activities through Discord. All platforms will be evaluated for accessibility before adoption.

Session 2 involves panel discussions and group activities. It starts with an *expert panel discussion* with invited speakers from diverse disciplines that contribute to XAI. Potentially, there could be a *short presentation session* to accommodate the remaining presentations. Then, *group discussion* takes place. Discussion topics will be crowd-sourced (via surveys prior to the workshop) and curated by the organizing committee. Teams are split into breakout rooms on Discord or sit together on-site (max 6 people per room). We have done a range of group activities in the past - from groups brainstorming “*papers from the future*” with thought-provoking ideas to “*news headlines in the near and far future*” where participants engage in design fiction and envision future issues or opportunities with XAI. These activities have led to many collaborations and papers from the participants. *After the group activity*, the participants regroup to share their discussions with quick “2-minute lightning talks”. In the *closing ceremony*, we wrap up the workshop with a short presentation summarizing the work and recognizing *impactful* position papers submitted. We also highlight areas of future work and propose ways to keep engaged with the HCXAI community through Discord and beyond.

Post Workshop Plans: Publications & Community Building. Following our plans of gradually transforming this workshop into its own conference series, we aim to fully publish submitted workshop papers as proceedings, using the EU open research platform Zenodo (<https://zenodo.org/>). As in previous installments, all additional workshop contents (in particular, videos of the keynotes, presentations, and main events) will be made accessible via our website.

Table 1: Tentative workshop structure, suggesting two 4-hour sessions (including breaks), as well as asynchronous activities before and after the workshop.

Start	End	Duration	Session
<i>Before the Workshop</i>			
-	-	2 weeks	Participants introduce themselves in the Discord channel and have access to provided workshop-related materials
<i>Workshop Session 1: 09:00–13:00 Japan Local Time</i>			
09:00	09:30	30min	Introduction of workshop organizers, topics, and goals
09:30	10:30	60min	Networking and Participant Introductions
<i>30 min break</i>			
11:00	12:00	60min	Keynote by invited speaker, including discussions
12:00	12:30	30min	Position paper session
12:30	13:00	30min	Poster session
<i>1 hour break</i>			
<i>Workshop Session 2: 14:00–18:00 Japan Local Time</i>			
14:00	15:30	90min	Panel presentations and panel discussion
<i>10 min break</i>			
15:40	17:10	90min	Break-out group work
<i>10 min break</i>			
17:20	17:45	25min	Break-out group findings presentations
17:45	18:00	15min	Closing ceremony & Wrap-Up
<i>After the workshop</i>			
-	-	-	Results summary posted on workshop website & initiating follow-up activities

6 Organizers

The Organizing Committee is uniquely positioned to execute the visions of the workshop. We are a global team spanning industry and academia and bridging relevant XAI threads like AI, HCI, Sociology, Civic Tech, Public Policy, and Psychology. Beyond hosting previous versions of this workshop, we have extensive organizational experience in HCI and AI venues.

Upol Ehsan is a Harvard Berkman Klein Fellow and incoming Assistant Professor of Computer Science at Northeastern. His work pioneered the domain of Human-centered XAI, receiving multiple awards at ACM CHI and HCII. Alongside serving as ACs in HCI and AI conferences (e.g., DIS, IUI, NeurIPS) and being the lead editor of the TiiS Journal Issue on HCXAI, he has been the lead organizer for the 2021, 2022, 2023, 2024 CHI workshops on Human-centred XAI at CHI.

Elizabeth Anne Watkins is a Sociotechnical Research Scientist at Intel Labs Intelligent Systems Research, where she serves on the Responsible AI Advisory Council. She was a Postdoctoral Fellow at Princeton University, with dual appointments at the Center for Information Technology Policy and the Human-Computer Interaction group. She has published or presented her research at CSCW, CHI, FAccT, USENIX, and AIES, and co-organized multiple CHI workshops including the 2022, 2023, and 2024 HCXAI workshops.

Philipp Wintersberger is a Full Professor of Intelligent User Interfaces at the Interdisciplinary Transformation University in Linz, Austria. His research addresses human-machine cooperation in safety-critical AI-driven systems. He is the founding chair of ACM SIGCHI Austria, and his research contributions have received several awards from venues such as ACM CHI, AutomotiveUI, and Interact.

Carina Manger is a researcher at the research center CARISSMA/THI. Before joining the Human-Computer Interaction Group, she obtained degrees in Psychology and Human Factors Engineering and worked on intelligent user interfaces in the automotive industry. Her current research concerns experimental user studies in simulated environments, with a strong focus on AI Explanations in automated driving.

Nina Hubig is an Assistant Professor at IT:U, the interdisciplinary technical University in Linz for XAI. She was prior an Assistant Professor at Clemson University with the Biomedical Data Science program, and holds an adjunct position with the Medical University of South Carolina (MUSC) and DePaul. Her research focuses on the use and creation of XAI methodologies in medicine to enhance legitimate trust, transparency and usability.

Saiph Savage is an Assistant Professor at Northeastern University and research collaborator at the Universidad Nacional Autonoma de Mexico (UNAM). She is an expert member of the Global Partnership on AI (GPAI), advising federal governments in the global south on integrating AI into national workforces. Recipient of an NSF CAREER award, she brings expertise around Civic AI and AI for policymakers in the Global South to the workshop.

Justin D. Weisz is a Research Manager at IBM Research in Yorktown Heights, NY. He is a co-lead of the Human-Centered Trustworthy AI strategy and his work explores the design of effective and safe user experiences with generative AI in enterprise settings. He has co-organized the HAI-GEN workshop series at IUI (2021+) and the HCXAI (2023) and ToMinHAI (2024) workshops at CHI. Dr. Weisz is an ACM Senior Member.

Andreas Riemer is a Professor for Human-Machine Interaction and VR at Technische Hochschule Ingolstadt with co-affiliation at

the CARISMA Institute of Automated Driving. His research interests include HF/ergonomics, adaptive UIs, driver state assessment, and trust/acceptance/ethics in mobility applications. Andreas is on the SC of ACM AutomotiveUI, chair of the German ACM SIGCHI chapter, and SC chair of “Mensch und Computer” (the German CHI).

7 Call for Participation

Explainable AI(XAI) is an essential pillar of Responsible AI. Explanations can improve real-world efficacy, provide harm mitigation levers, and serve as a primary means to ensure humans’ right to understand and contest decisions made about them by AI systems. This workshop serves as a junction point of cross-disciplinary stakeholders of the XAI landscape to examine how human-centered perspectives in XAI can be operationalized at the conceptual, methodological, and technical levels. We call for *position or empirical papers* making justifiable arguments that address topics involving the *who* (e.g., relevant diverse stakeholders), *why* (e.g., social/individual factors influencing explainability goals), *when* (e.g., when to trust the AI’s explanations vs. not) or *where* (e.g., diverse application areas, XAI for actionability or human-AI collaboration, or XAI evaluation).

In the *fifth* Human-centered XAI workshop, there is a special focus on emerging frontiers of XAI: (1) participatory approaches toward explainability in civic AI applications; (2) addressing hallucinations in LLMs using explainability benchmarks; (3) connecting HCXAI research with Responsible AI practices, algorithmic auditing, and public policy; and (4) improving representation of XAI issues from the Global South.

Papers should follow the CHI Extended Abstract format (<=4 pages excluding references) and be submitted through the workshop’s submission site (<https://hcxai.jimdosite.com/>). Reviews are author-disclosed (single-blind). All accepted papers will be presented, provided at least one author registers for the entirety of the workshop. We especially encourage participation from the Global South and underrepresented stakeholders in the dominant XAI discourse, striving for a more equitable conversation.

References

- [1] Ashraf Abdul, Jo Vermeulen, Danding Wang, Brian Y Lim, and Mohan Kankanhalli. 2018. Trends and trajectories for explainable, accountable and intelligible systems: An hci research agenda. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–18.
- [2] Alejandro Barredo Arrieta, Natalia Díaz-Rodríguez, Javier Del Ser, Adrien Ben- netot, Siham Tabik, Alberto Barbado, Salvador García, Sergio Gil-López, Daniel Molina, Richard Benjamins, et al. 2020. Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion* 58 (2020), 82–115.
- [3] Zoe M. Becerra, Nadia Fereydooni, Andrew L. Kun, Angus McKerral, Andreas Riener, Clemens Schartmüller, Bruce N. Walker, and Philipp Wintersberger. 2021. Interactive Workshops in a Pandemic: The Real Benefits of Virtual Spaces. *IEEE Pervasive Computing* 20, 1 (2021), 35–39. <https://doi.org/10.1109/MPRV.2020.3044072>
- [4] Joseph R Biden. 2023. Executive order on the safe, secure, and trustworthy development and use of artificial intelligence. (2023).
- [5] Wiebe E Bijker, Thomas P Hughes, Trevor Pinch, et al. 1987. The social construction of technological systems.
- [6] Upol Ehsan, Q Vera Liao, Samir Passi, Mark O Riedl, and Hal Daume III. 2022. Seamless XAI: Operationalizing Seamless Design in Explainable AI. *arXiv preprint arXiv:2211.06753* (2022).
- [7] Upol Ehsan and Mark O Riedl. 2020. Human-centered Explainable AI: Towards a Reflective Sociotechnical Approach. *arXiv preprint arXiv:2002.01092* (2020).
- [8] Upol Ehsan and Mark O Riedl. 2021. Explainability pitfalls: Beyond dark patterns in explainable AI. *arXiv preprint arXiv:2109.12480* (2021).
- [9] Upol Ehsan, Pradyumna Tambwekar, Larry Chan, Brent Harrison, and Mark O Riedl. 2019. Automated rationale generation: a technique for explainable AI and its effects on human perceptions. In *Proceedings of the 24th International Conference on Intelligent User Interfaces*. 263–274.
- [10] Upol Ehsan, Elizabeth A Watkins, Philipp Wintersberger, Carina Manger, Sun- nie SY Kim, Niels Van Berkel, Andreas Riener, and Mark O Riedl. 2024. Human-Centered Explainable AI (HCXAI): Reloading Explainability in the Era of Large Language Models (LLMs). In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–6.
- [11] Upol Ehsan, Philipp Wintersberger, Q Vera Liao, Martina Mara, Marc Streit, Sandra Wachter, Andreas Riener, and Mark O Riedl. 2021. Operationalizing Human-Centered Perspectives in Explainable AI. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [12] Upol Ehsan, Philipp Wintersberger, Q Vera Liao, Elizabeth Anne Watkins, Carina Manger, Hal Daumé III, Andreas Riener, and Mark O Riedl. 2022. Human-Centered Explainable AI (HCXAI): Beyond Opening the Black-Box of AI. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–7.
- [13] Upol Ehsan, Philipp Wintersberger, Elizabeth A Watkins, Carina Manger, Gonzalo Ramos, Justin D Weisz, Hal Daumé III, Andreas Riener, and Mark O Riedl. 2023. Human-Centered Explainable AI (HCXAI): Coming of Age. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–7.
- [14] D Gunning. 2016. Explainable artificial intelligence (XAI). Defense Advanced Research Projects agency.
- [15] Q Vera Liao and Kush R Varshney. 2021. Human-centered explainable ai (xai): From algorithms to user experiences. *arXiv preprint arXiv:2110.10790* (2021).
- [16] Tambiama Madiega. 2021. Artificial intelligence act. *European Parliament: European Parliamentary Research Service* (2021).
- [17] Sina Mohseni, Niloofar Zarei, and Eric D Ragan. 2018. A Multidisciplinary Survey and Framework for Design and Evaluation of Explainable AI Systems. *arXiv* (2018), arXiv–1811.
- [18] Paul Schmitt, Swapna Joshi, Bruce N Walker, and Azra Habibovic. 2024. Symbiotic-A Workshop on Transferability between Human-Robot Interaction (HRI) and Autonomous Vehicle Interaction Research. In *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. 273–276.
- [19] William R Swartout. 1983. XPLAIN: A system for creating and explaining expert consulting programs. *Artificial intelligence* 21, 3 (1983), 285–325.
- [20] Philipp Wintersberger, Hannah Nicklas, Thomas Martlbauer, Stephan Hammer, and Andreas Riener. 2020. Explainable Automation: Personalized and Adaptive UIs to Foster Trust and Understanding of Driving Automation Systems. In *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Virtual Event, DC, USA) (*AutomotiveUI ’20*). Association for Computing Machinery, New York, NY, USA, 252–261. <https://doi.org/10.1145/3409120.3410659>